

units of D are s^{-1} . Hence D has unit s^{-1} .

2 Ans: B

$$\rho = mL / (t_2 - t_1) = (52.000)(10.0) / (0.63 - 0.00) = 825.397 \text{ g cm s}^{-1}$$

$$\frac{\Delta p}{p} = \frac{\Delta m}{m} + \frac{\Delta L}{L} + \frac{\Delta(t_2 - t_1)}{(t_2 - t_1)}$$

$$= \frac{0.001}{52} + \frac{0.1}{10} + \frac{0.02}{0.63}$$

$$= 0.041765$$

$$\Delta p = 34.5 \text{ g cm s}^{-1} = 30 \text{ g cm s}^{-1} \text{ (1 s.f.)}, \text{ \{so since } \Delta p \text{ is now expressed in 'tens', } \\ p \text{ must be expressed to nearest 'tens', or 830 \}}$$

$$\rho = 830 \pm 30 \text{ g cm s}^{-1}$$

3 Ans: B